Use of OpenRefine

Biodiversity Data Mobilization - Data Cleaning - OpenRefine Exercise

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CONVENTIONS

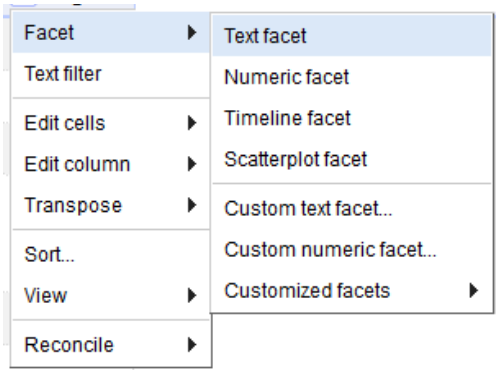
Formulas (copy-paste) `Text in blue`

Example: ...then paste the expression `^[a-z]`



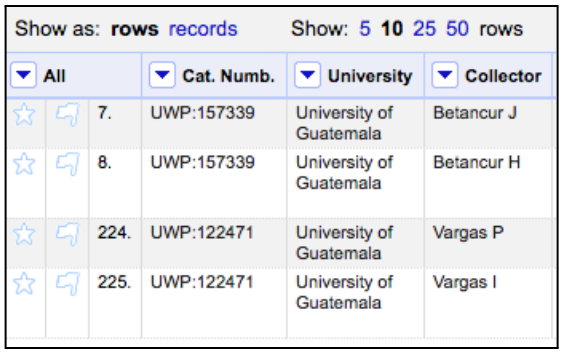
Commands in Refine Text in red

Example: ...and follow the route to Text facet



Column names Text in green

Example: ...go to column Cat. Numb



Hyperlinks www.gbif.org

Column menu 

2. BASIC USE

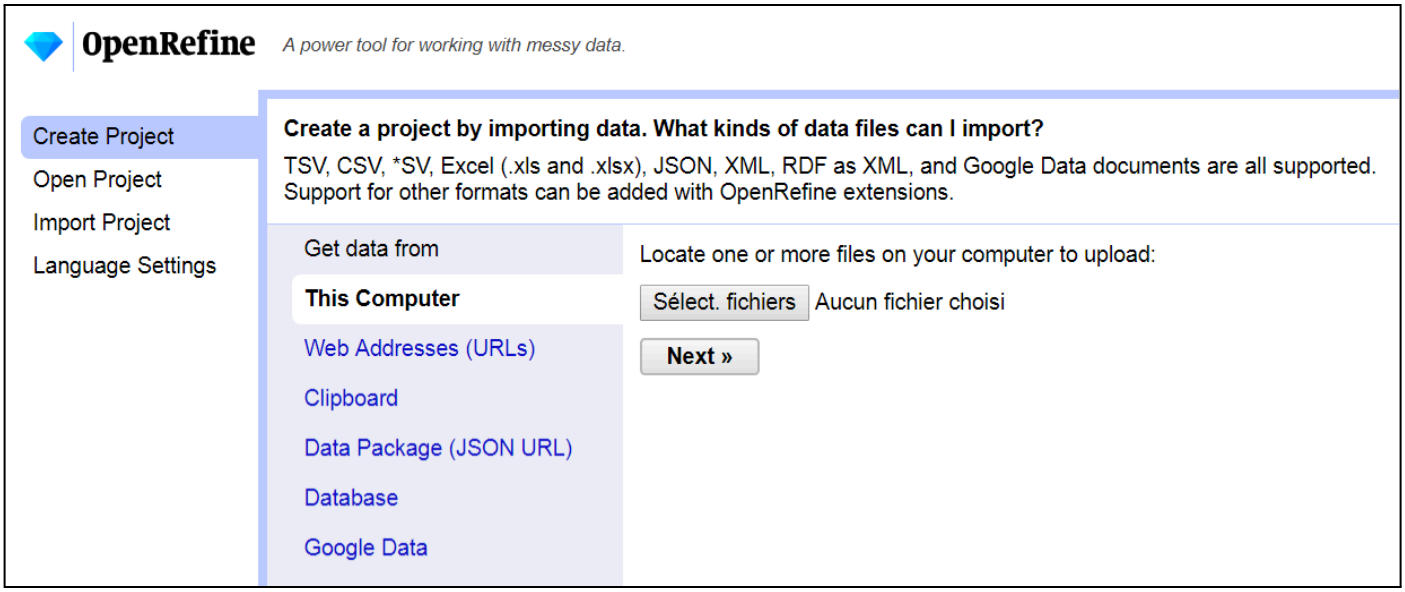
2.1. FILE LOADING AND PROJECTS

2.1.1. Before starting

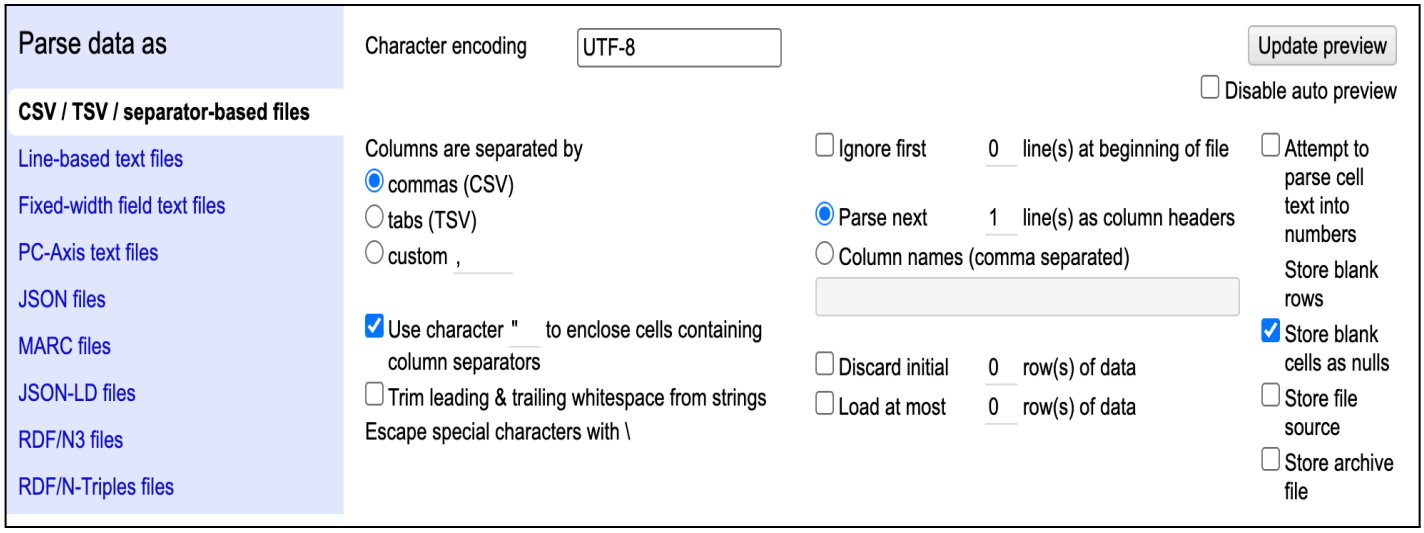
Data loading can be done from various data sources: TSV, CSV, SV, Excel (.xls and .xlsx), JSON, XML, RDF and XML data as Google Docs. Loading data involves two stages: the first is the file load and the second is the creation of the project.

2.1.2. Exercise 1. Create a project

1. Load the data file from the indicated link one the training site.
2. Open *OpenRefine* (in the latest version of OpenRefine, you can select your preferred language in Language Settings), click on Create Project, and follow the route Get data from > This Computer, then click on Choose Files. Select the file. Click on Next.



3. OpenRefine will automatically detect the format of your file as csv; a parsing options menu will appear. Choose the options as shown in the picture:



4. On the top right in the Project Name field, rename your file as [YourName]UC-Practice-OpenRefine and click Create Project and you will be ready to work!
5. You will only see 10 rows to start. You can show greater amounts of records if you wish.

304 rows Extensions: Wikibase

Show as: **rows** records Show: 5 **10** 25 50 100 500 1000 rows « first ‹ previous 1 next › last »

All		occurrenceID	decimalLatitude	decimalLongitude	country	countryCode	county	recordedBy	scientificName	Api_name
1.		SBN:4039:201059			Sweden	SE	Hallands län	SBN:recorderId:300	Aglais io	
2.		SBN:4039:201062			Sweden	SE	Hallands län	SBN:recorderId:300	Aglais urticae	
3.		SBN:4039:201044			Sweden	SE	Hallands län	SBN:recorderId:300	Anthocharis cardamines	
4.		SBN:4039:201107			Sweden	SE	Hallands län	SBN:recorderId:300	Aphantopus hyperantus	
5.		SBN:4039:201038			Sweden	SE	Hallands län	SBN:recorderId:300	Aporia crataegi	
6.		SBN:4039:201064			Sweden	SE	Hallands län	SBN:recorderId:300	Araschnia levana	
7.		SBN:4039:201065			Sweden	SE	Hallands län	SBN:recorderId:300	Argynnis paphia	
8.		SBN:4039:201072			Sweden	SE	Hallands län	SBN:recorderId:300	Boloria aquilonaris	
9.		SBN:4039:201071			Sweden	SE	Hallands län	SBN:recorderId:300	Brenthis ino	
10.		SBN:4039:201137			Sweden	SE	Hallands län	SBN:recorderId:300	Celastrina argiolus	

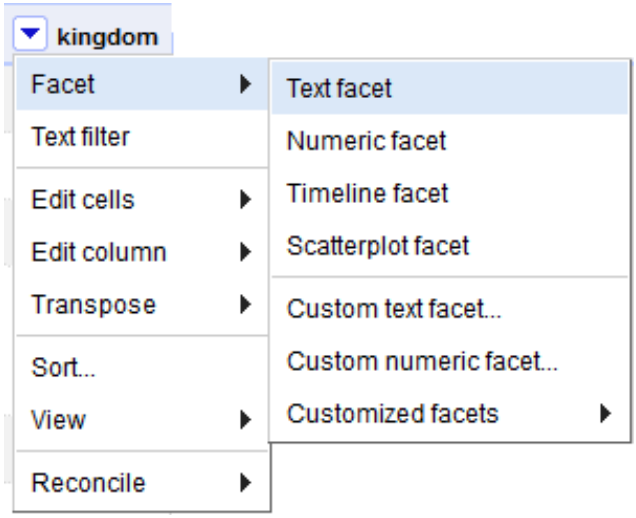
2.2. FACETING

2.2.1. Before starting

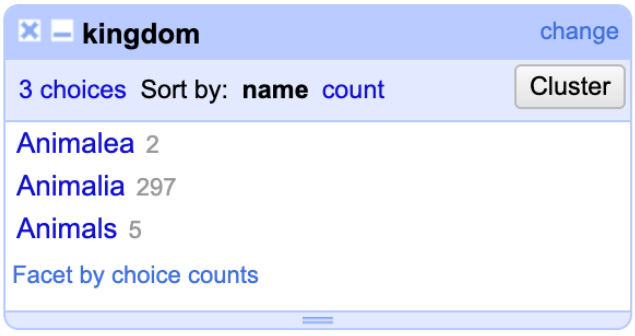
Faceting is a feature that will allow us to get a big picture overview of the data, and to filter down to just the subset of rows that we want to change or view in bulk. It facilitates the use and analysis of data and can be done with cells containing any kind of text, numbers and dates. After performing any type of facet, you may wish to close it when you are done.

2.2.2. Exercise 2. Faceting and mass editing

1. Go to column kingdom, and then click on the column menu and follow the route to Text facet as shown below:

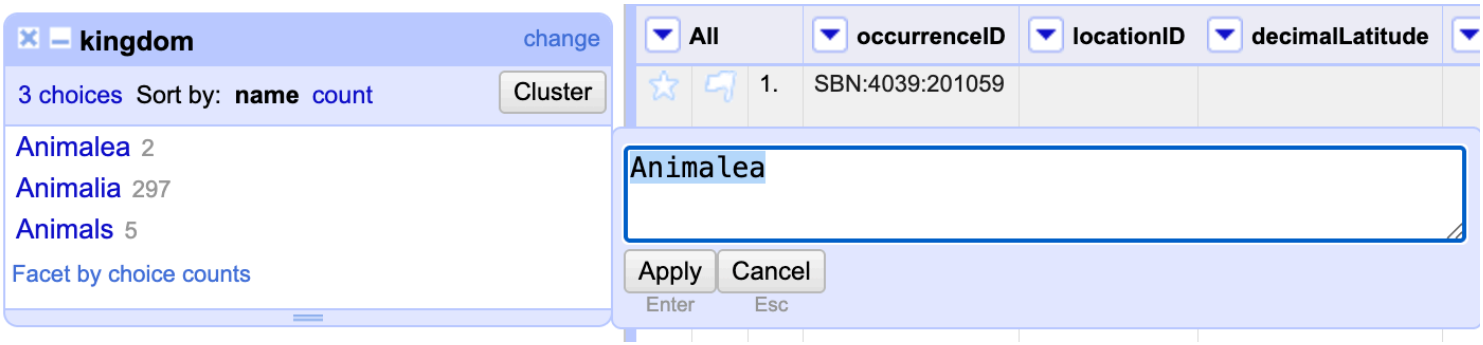


2. On the left a window with the name of the column will appear, that is the facet:



   Click on count to sort by count, then click on name to sort alphabetically.

3. Fix the spelling mistakes. Place the cursor over the text in the window and click on edit, then fix the error in the text box, and to save click on apply.



All the values will be fixed automatically.

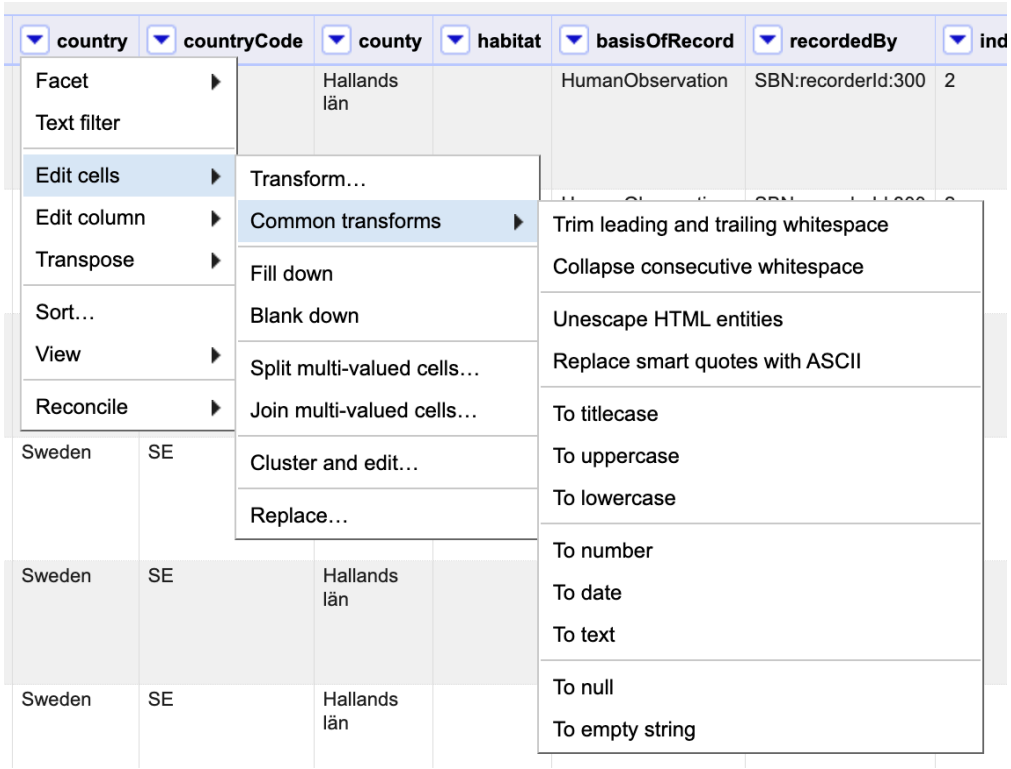
2.2.3. Exercise 3. Faceting and white spaces I

1. Go to country and click on column menu  and perform a Text Facet.

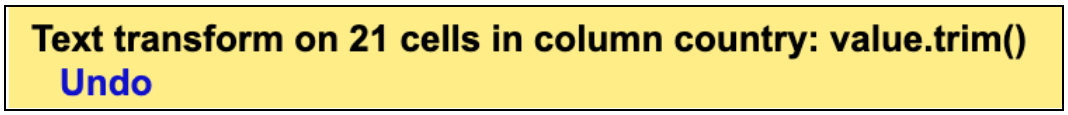


On a quick view, the country appears to be spelled correctly, but the facet shows three different values due to the extra spaces at the end of the text.

2. Fix the error from the column menu on country., following the route Edit Cells > Common transforms > Trim leading and trailing whitespace.



You will see a notification message:



3. Now check the facet window; only one value will remain.

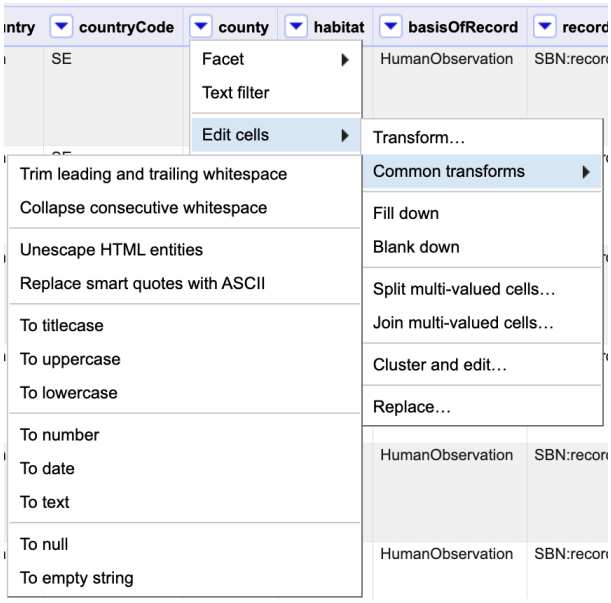


2.2.4. Exercise 4. Faceting and white spaces II

1. Go to column county and click on  then go to Text facet. Then click on count. The facet will show:



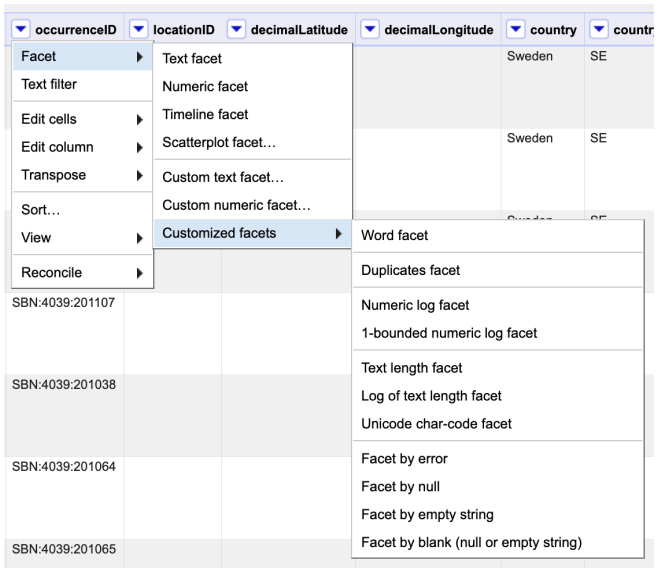
As seen above, *Skåne län* is the first item in the list with 57 specimens, but it is also present in the 3rd place with 37 specimens.



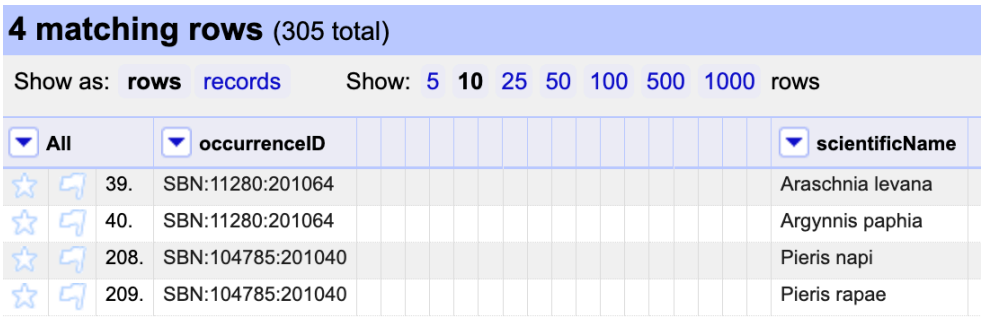
2. Fix the error from the county column menu, Edit Cells > Common transforms > Collapse consecutive whitespaces.
3. Once the white-spaces are removed, *Skåne län* should only appear in the list with 94 records.

2.2.5. *Exercise 5. Faceting and duplicates*

1. Go to column catalog in occurrenceID, and follow the route Facet > Customized facets > Duplicates facet. The facet will show 4 duplicates.



2. Click on true, and you'll see the values in the main window:



3.

4. After a check with the transcribers, you are now able to match the scientificNames with their correct occurrenceID. Edit the values directly on the cell.

```
SBN:11280:201064      Araschnia levana

SBN:11280:201065      Argynnis paphia

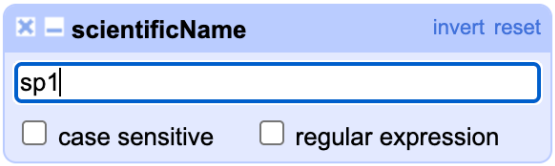
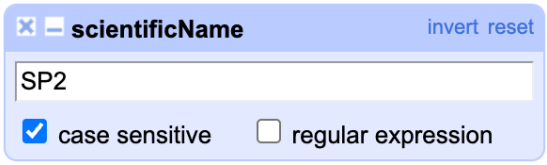
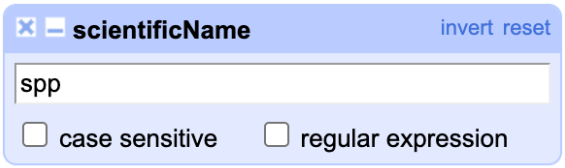
SBN:104785:201041     Pieris napi

SBN:104785:201040     Pieris rapae
```

2.3. FILTERING

2.3.1. Exercise 6. Basic filter

1. Go again to scientificName column menu and perform a Text facet to visualize the values, then go again to ▾ and click on Text filter, perform the following filters and fix them as shown below:

Filter	How to fix	Correct value
	Edit directly in the cell	Hesperiidae Pieridae Polyommatini
	Edit directly in the cell, check case sensitive	Polyommatini
	1. Go to ▾ on scientificName, then click Edit cells > Transform... 2. In the text box paste the formula `value.replace(" spp", "")` 3. Click OK	Heliconiinae Zygaenidae Pieridae

2.3.2. Exercise 7. Advanced filter I

1. Go to column genus and perform a Text filter.
2. Check regular expression and case sensitive, then paste the expression `^[a-z]`



   This regular expression filters the strings in which the first letter is lowercase.

3. Perform a correction since the genus should be capitalized.

Note: If you want to know more about regular expressions click here.

2.3.3. Exercise 8. Advanced filter II

1. Go to column scientificName and perform a Text filter.
2. Check regular expression and case sensitive, then paste the expression `^[A-Z].*\s[A-Z]`



   This regular expression filters the strings that start with a capital letter followed by 0 or more characters, then a space, then a capital letter.

3. Perform a correction since the second word of the name should be lowercase.

Note: If you want to know more about regular expressions click here.

2.4. CLUSTERING

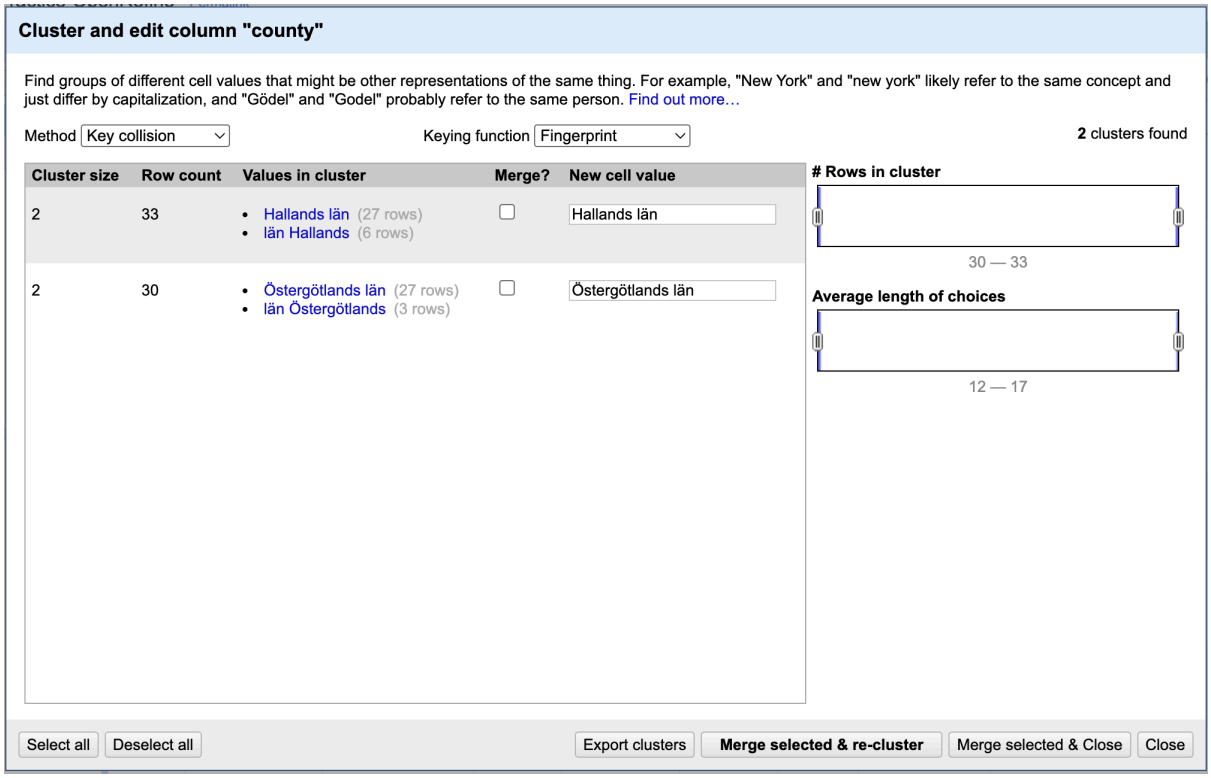
2.4.1. Exercise 9. Basic clustering

1. Go to county, then in the menu column click Text facet.



Keep in mind that the correct counties are: Hallands län, Jönköpings län, Östergötlands län, Skåne län, Stockholms län, Värmlands län, Västmanlands län.

2. On the top right of the facet window click on Cluster, a new window will appear:



3. Now you can see information about the clusters:
    - Cluster size: the number of different versions that the clustering algorithm believes to be the same.
    - Row count: the number of records with any of the cluster values.
    - Values in cluster: the actual values that the algorithm believes to be the same. There is also the number of records with each particular value, and the possibility to browse the contents of the cluster in a different tab.
    - Merge?: check if values are to be merged into a single standard value.
    - New cell value: the value to be applied to every record in the cluster. By default, it is the value with most records. You can also click on any value to apply that to the New cell value.

    Note: If you want to know more about clustering click here.

4. Click on Select All and then on Merge Selected & close, you will see a notification message:



5. To fix the remaining counties go again to Cluster in the facet window of county.
6. In the Cluster and edit window, go to Keying Function, then select ngram-fingerprint, and set 1 as the value in Ngram Size. Press the enter key.
7. Click on Select All and then on Merge Selected & close, you will see a notification message:



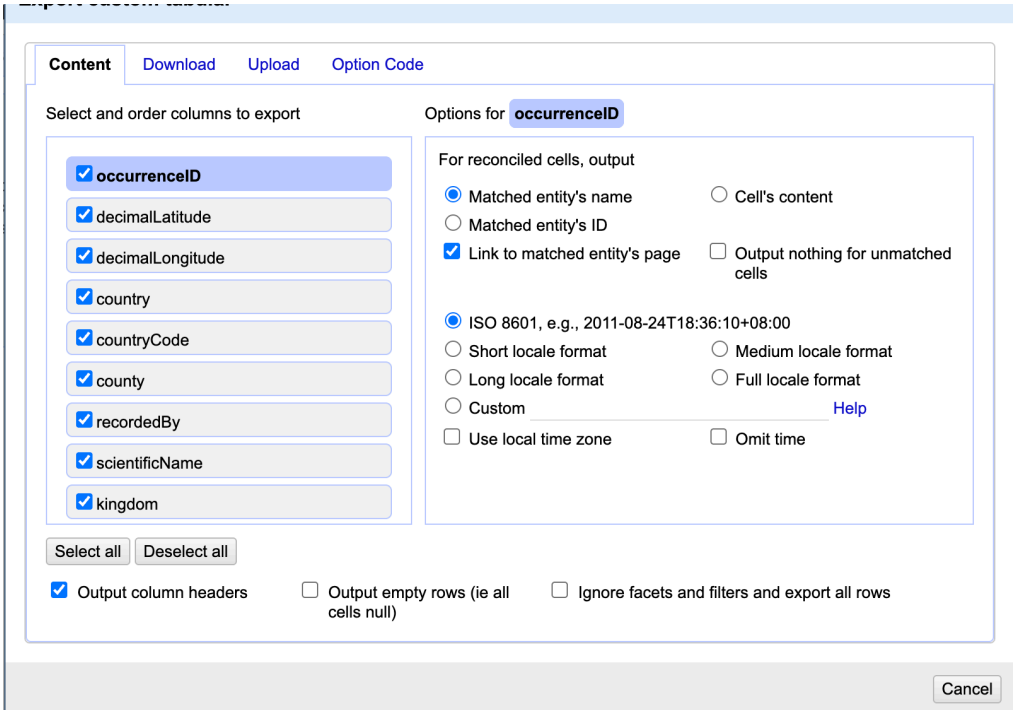
8. Your counties are now fixed and your window should look like the image below:



2.5. EXPORTING

You will have several options for exporting your cleaned data, but the following option is useful in most cases.

1. On the upper right corner click on Export and select Custom tabular ...
2. You will see the exporter window:



3. In the content tab you can choose the columns that you want to export, if you select Ignore facets and filters and export all rows all facets and filterings will be ignored, this is useful if you forget to clear them before exporting.
4. Go to the Download tab and select the separator that you prefer. Don't modify the other options unless you need to.

You can also export the whole project to open it in OpenRefine on another computer by following the route Export > Export project. In this case you're not downloading a data file to open in a spreadsheet or text processor, but rather a GZIP file that will only be accessible through OpenRefine.

3. BASIC API USE

3.1. Before starting

Reconciliation matches the information in one of your columns to an outside database. This is particularly helpful when it comes to name validation, as it proves the name you have exists somewhere else. This is a really useful service, but can be time consuming. In this case we will go through the process with only three records using the API from GBIF. Internet connection is required.

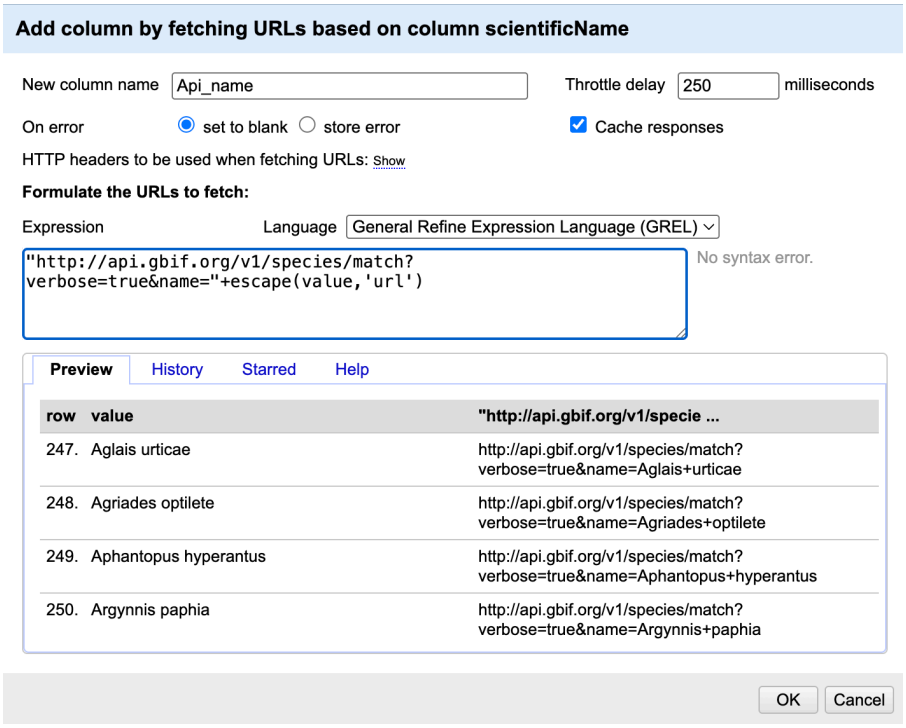
3.2. Exercise 1. Higher taxonomy

1. Go to recordedBy, then make a Text facet. Select SBN:recorderId:716



2. Under scientificName, click on column menu and then Edit column > Add column by fetching URLs..., call the new column Api_name
3. Change the Throttle Delay to `250` and paste the expression:

```
"http://api.gbif.org/v1/species/match?verbose=true&name="+escape(value,'url')
```

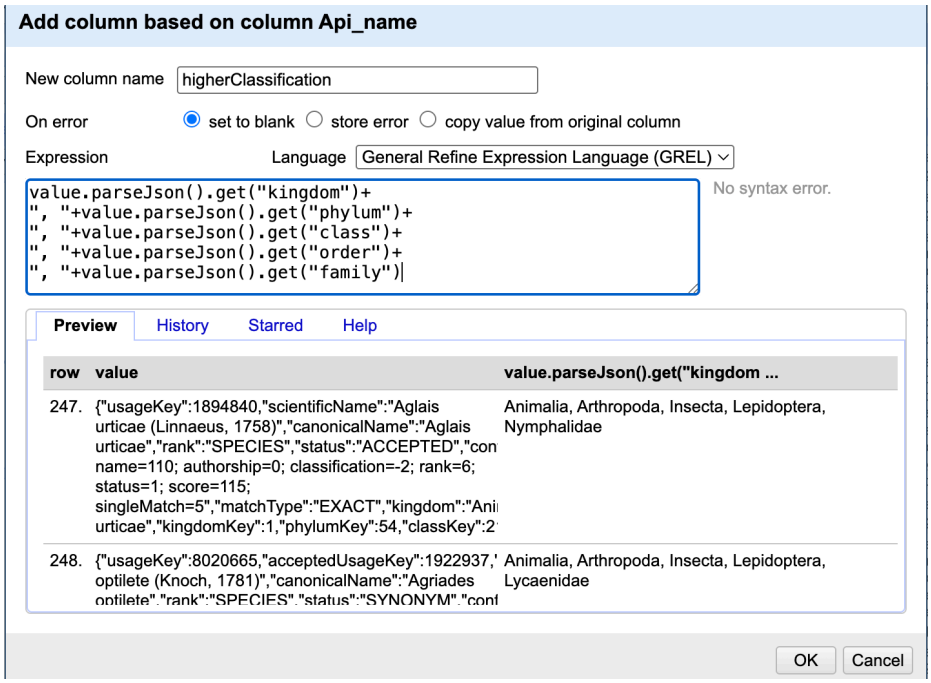


4. Click ok and wait, this might take some time depending on your internet connection and the number of taxa.

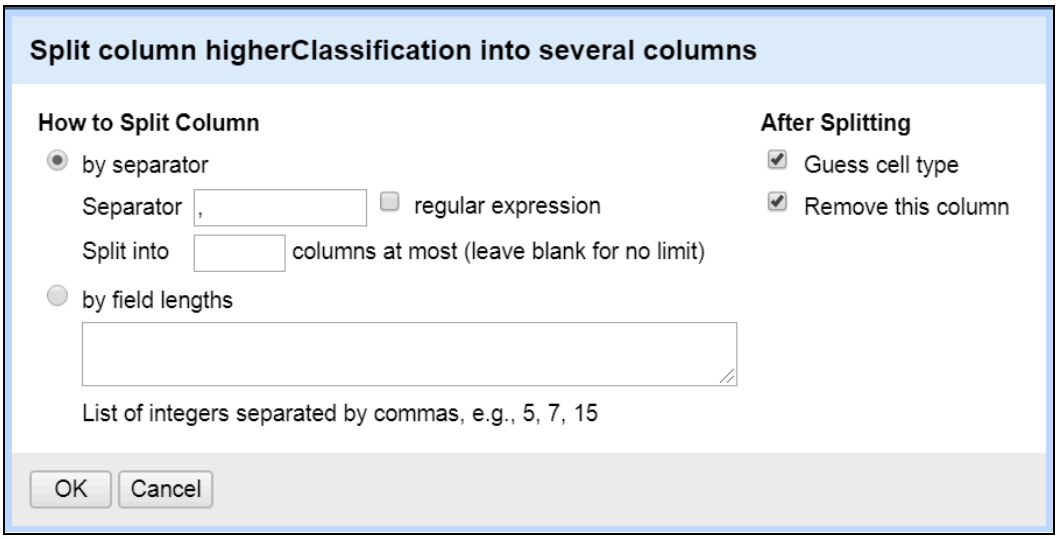
5. Go to Api_name, click on column menu and then Edit column > Add column based on this column.... Call the new column higherClassification and paste the expression:

```
value.parseJson().get("kingdom")+
", "+value.parseJson().get("phylum")+
", "+value.parseJson().get("class")+
", "+value.parseJson().get("order")+
", "+value.parseJson().get("family")
```

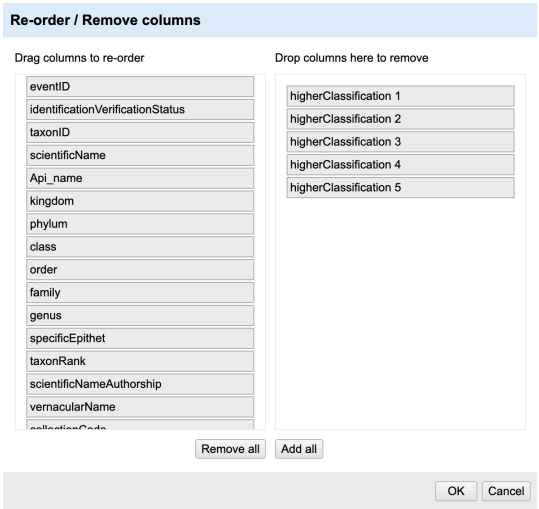
You will see the Kingdom, Phylum, Class, Order and family of each taxon. Click OK.



6. Under higherClassification follow the route Edit column > Split into several columns..., leave the settings as shown and click OK:



7. Now you know how to obtain the taxonomic categories of a given taxon if this is available in the GBIF API. Column names can be edited in Edit column > Rename this column.
8. For the purpose of the workshop, the columns created in this exercise (Higher taxonomy) must be deleted. Under All, which is the first column, go to Edit columns > Re-order / remove columns...
9. Drop the columns as shown and click OK:



4.USEFUL LINKS AND REFERENCES

- Name validation Tutorial: https://docs.google.com/document/d/1tkDRXlYhmassYAk5T4v5oac5prF0jAiSMr_JEGTvhRo/edit
- Higher Taxonomy Tutorial: https://docs.google.com/document/d/1XZ_pM9gIdQzHzl8wfUCVea-52yub5T_3tc-snBgPRa0/edit
- Documentation https://github.com/OpenRefine/OpenRefine/wiki/Documentation-For-Users
- Resources list for OpenRefine: https://github.com/OpenRefine/OpenRefine/wiki/External-Resource

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Exercise concept and content developed by Néstor Beltrán.
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